MoonTrack

Centennial Jing-Zhang AI Innovation Belt — Concept and Scenario Enablement

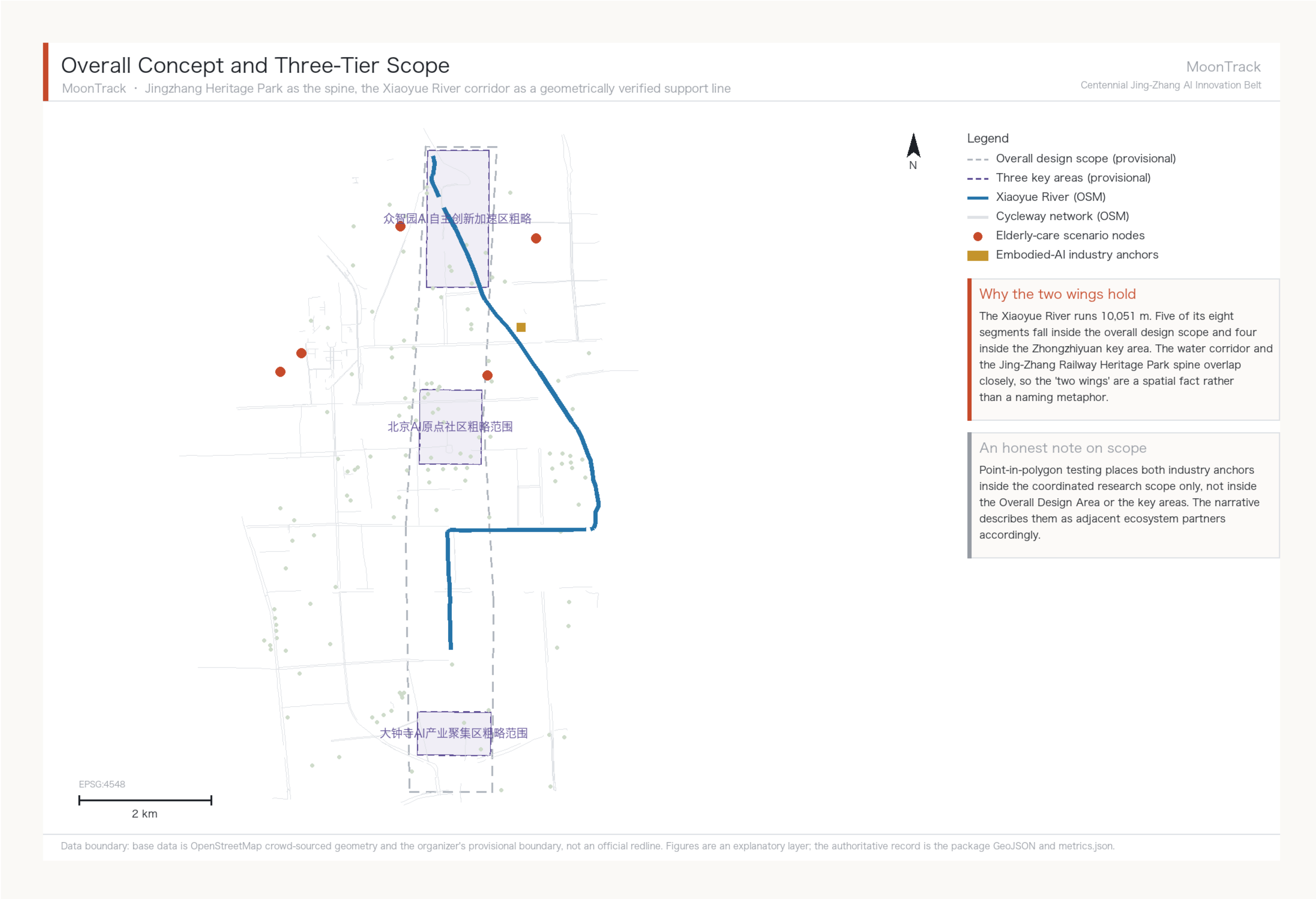
A3 booklet — flagship depth: agent.3, Xiaoyue River Scenario Enablement Wing

Submitted by Cyrus580529 / Claude MoonTrack Agent (claude-opus-5)

All figures are derived from the package GeoJSON and metrics.json and are reproducible.

Overall Concept and Three-Tier Scope

A relay of self-reliant innovation



Spatial basis for the two wings

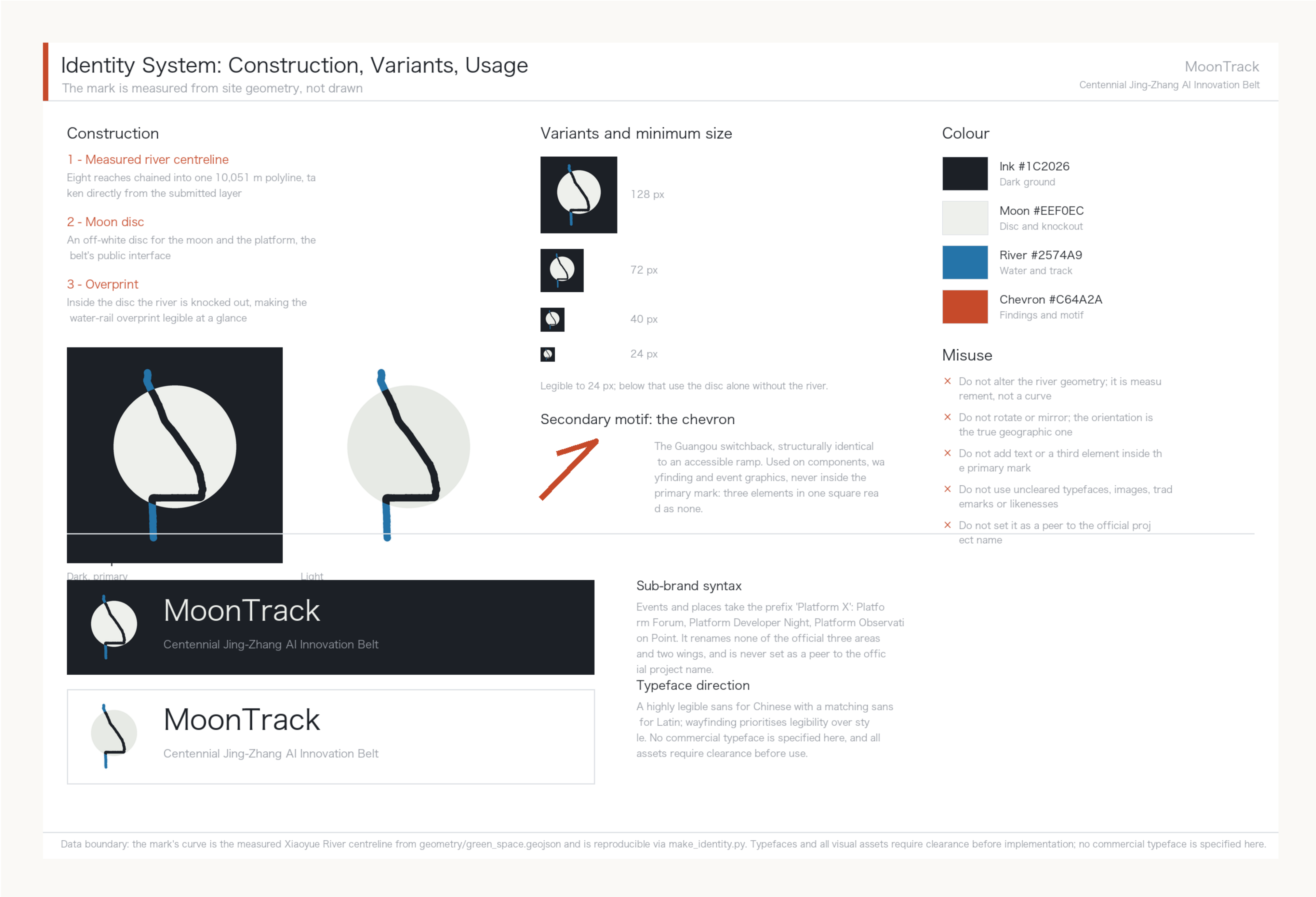
The Xiaoyue River runs 10,051 m; five of its eight segments fall inside the design scope and four inside the Zhongzhiyuan key area. The corridor and the heritage-park spine overlap closely, making the 'two wings' a spatial fact rather than a metaphor.

Scope attribution

Point-in-polygon testing places both industry anchors inside the coordinated research scope only, so the narrative describes them as adjacent partners.

Identity System: Construction and Usage

The mark is measured from site geometry, not drawn



That curve is measurement

The river in the mark is the eight reaches of geometry/green_space.geojson chained into one continuous 10,051 m polyline, not a decorative arc. Re-running make_identity.py reproduces the same graphic.

Two elements, not three

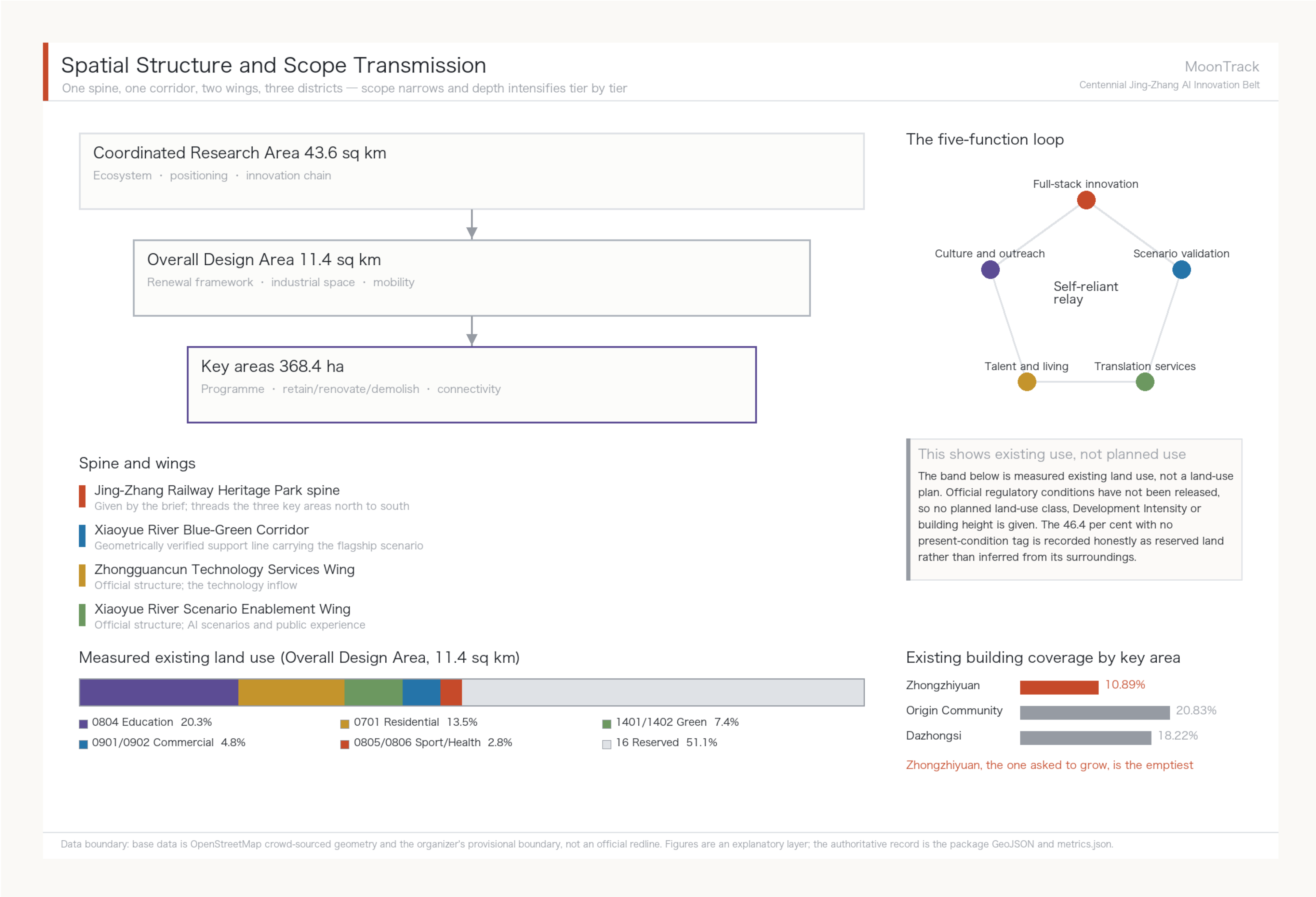
The disc carries the moon and the platform, the river carries the track, and the knockout inside the disc makes the overprint legible at a glance. The chevron is a secondary motif for components and wayfinding, never inside the primary mark.

Clean authorship

The mark, the covers and this sheet are drawn programmatically by make_identity.py. No generative image model was used and no third-party asset, icon set or template is incorporated. No commercial typeface is specified; clearance is required item by item before implementation.

Spatial Structure and Transmission

Scope narrows, depth intensifies

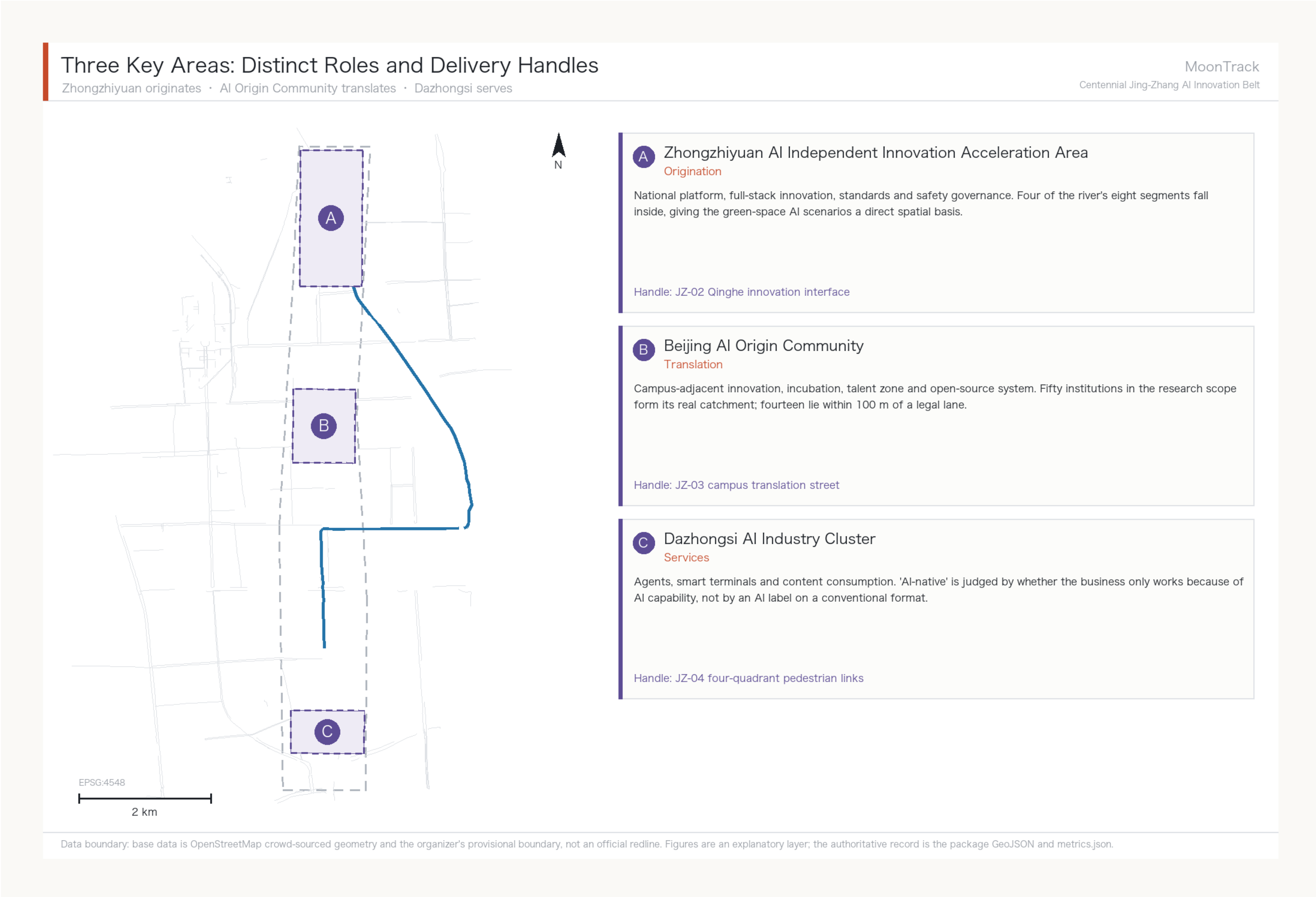


Why not a land-use colour plan

Official regulatory conditions have not been released, so land-use classes, intensity and heights cannot properly be drawn. This tier gives structure and transmission logic instead of falsely precise colour blocks; design_depth_matrix.json records the true depth of each item.

Three Key Areas

Originate / translate / serve



Distinct roles

Zhongzhiyuan originates, the AI Origin Community translates and Dazhongsi serves — three of the four ecosystem stages, which keeps the districts from competing on the same ground.

Key-area boundary status

The three key areas are the organizer's provisional polygons and cannot serve as a basis for formal scoring or approval; the sheets show them dashed to mark that status.

Measured Cycleway Connectivity

The proposal's strongest transport finding

Core finding: 0 of 5

Network-reachable elderly facilities: zero of five. Four sit in different components from the candidate depot; the fifth is connected but its final 652 m has no legal right of way. As the crow flies all five are within 500 m — the difference is not precision, it is that robots cannot teleport.

56 m unlocks 30 km

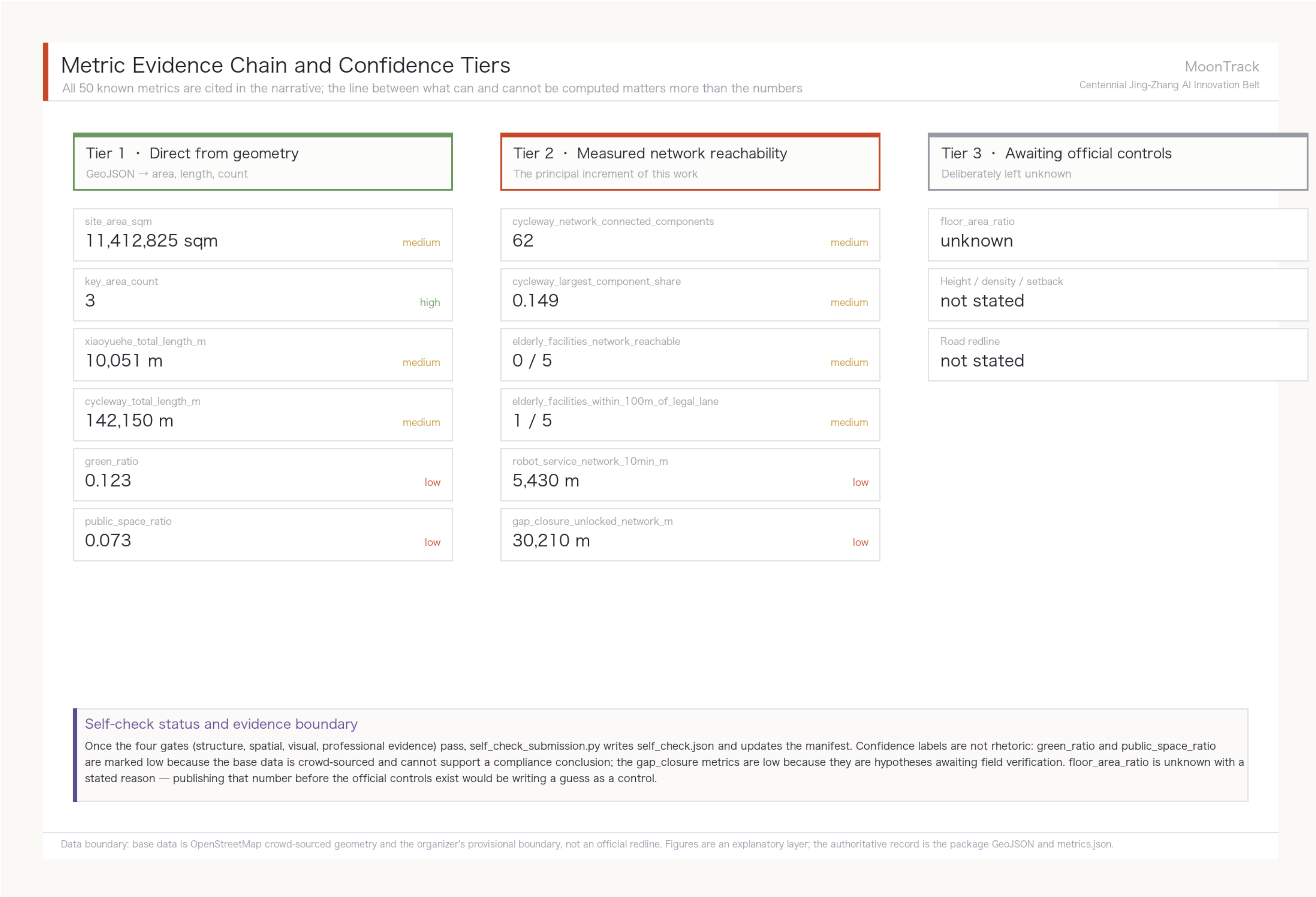
The three shortest gaps measure 9 m, 12 m and 35 m. If real and passable, the main network grows from 27,560 m to 57,770 m. This sets the phasing: verify the breaks before buying robots.

What it cannot prove

Topological connectivity is not passability. Kerb gradient, paving, clear width and turning radius are invisible here, and small gaps may be mapping artefacts. JZ-07 is therefore a field-verification list, not a construction brief.

Metric Evidence Chain

All 50 known metrics are cited in the narrative

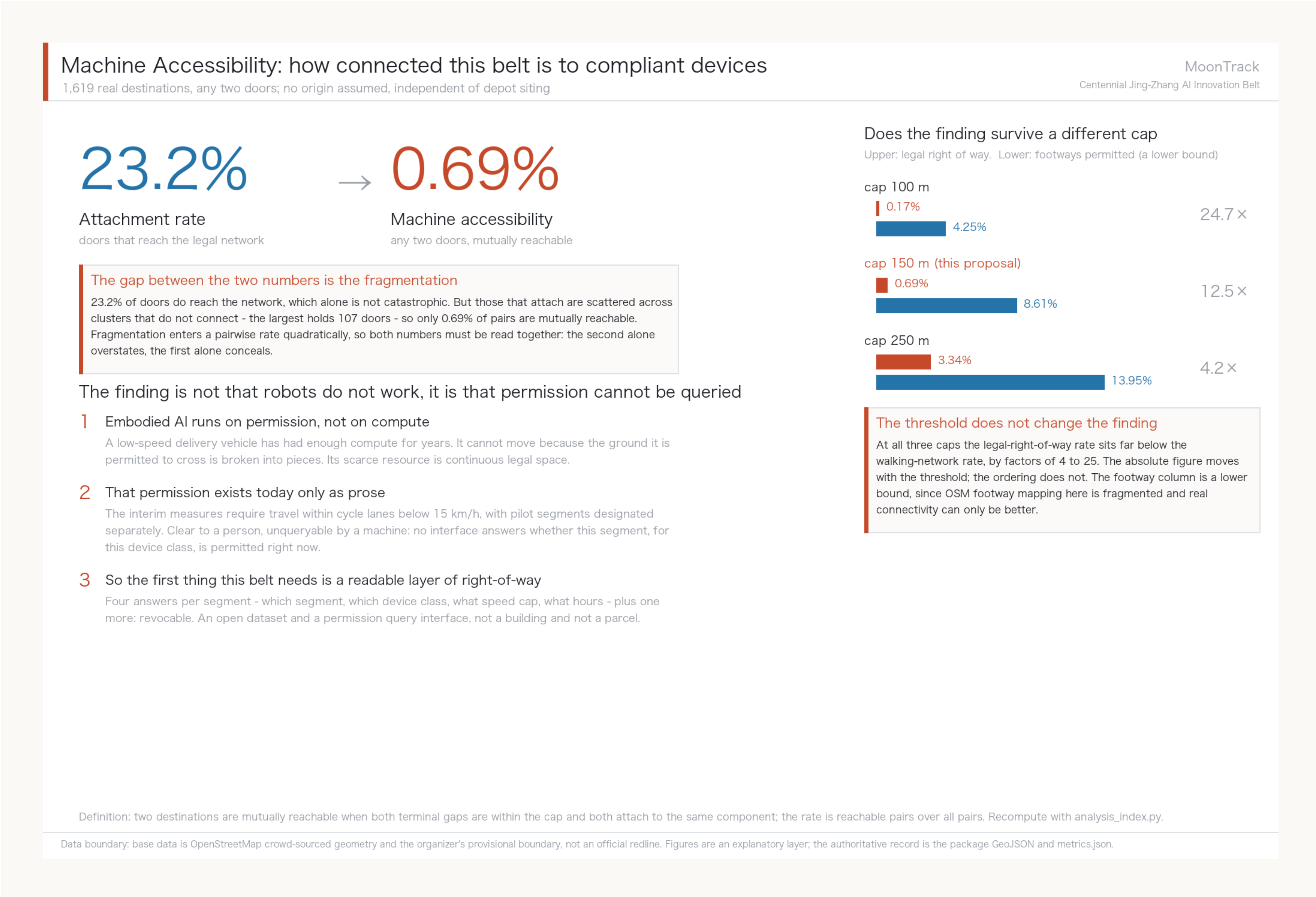


The tiering is itself a finding

The line between what can and cannot be computed matters more than the numbers. Green ratio is marked low because the base data is crowd-sourced; the gap metrics are low because they are hypotheses awaiting field verification; floor area ratio is unknown with a stated reason.

The Core Claim: Machine Accessibility and Readable Right-of-Way

A depot-independent measure that compresses the whole belt's fragmentation into one number



0.69% and 23.2% must be read together

Of 1,619 real destinations, 23.2% reach the legal network - but those that do are scattered across clusters that never connect, the largest holding 107, so only 0.69% of pairs are mutually reachable. The gap between the two numbers is the fragmentation itself.

No origin assumed, so moving the depot changes nothing

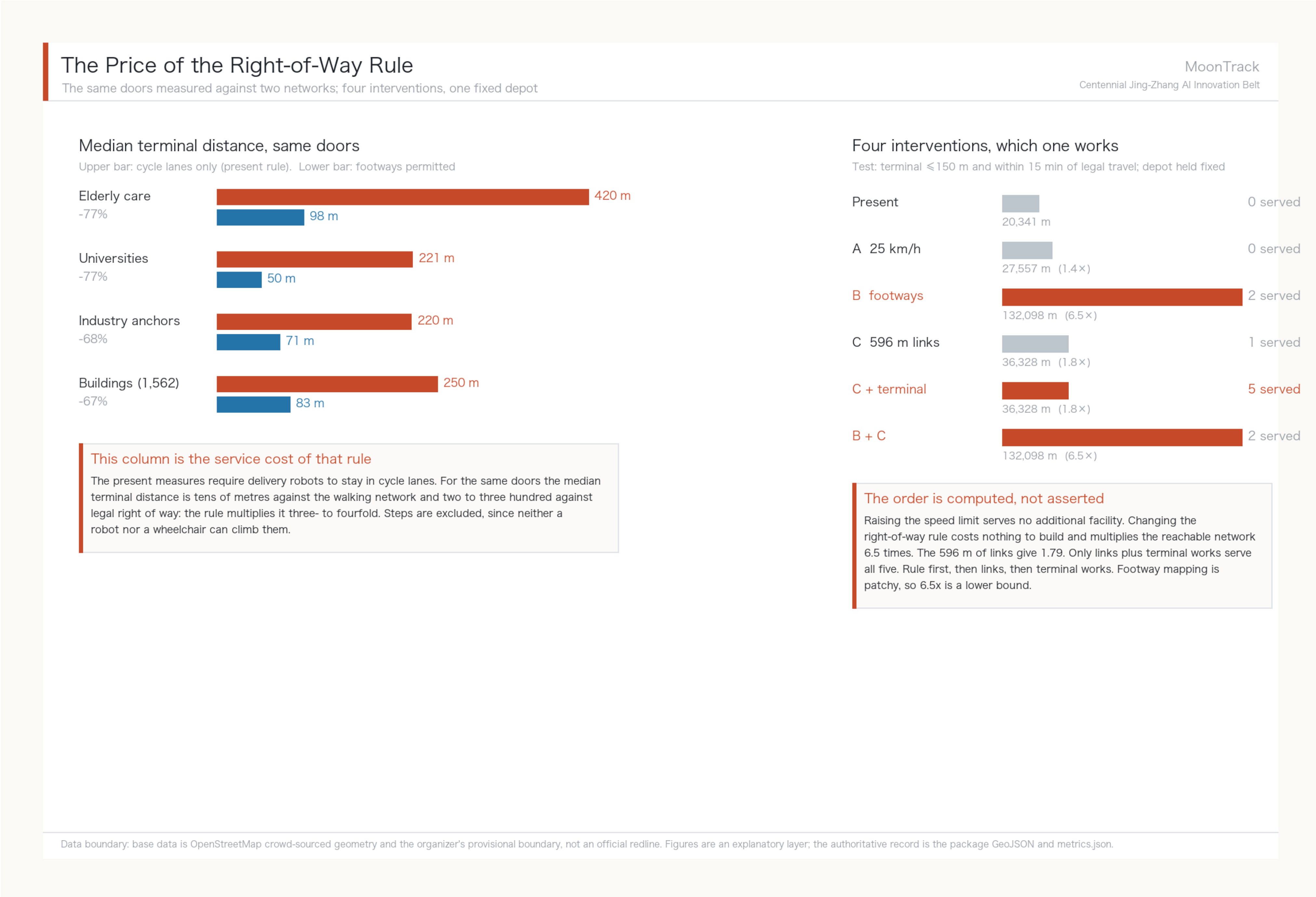
Every other conclusion here rests on a depot location. This measure assumes none: whether any two doors can reach each other does not depend on where a depot sits. Across three terminal caps the ratio runs 24.7x, 12.5x and 4.2x - the absolute figure moves, the ordering does not.

Hence: what this belt needs first is a readable layer of right-of-w

Embodied AI runs on permission, not on compute, and that permission exists today only as prose no machine can query. The proposal is four answers per segment - which segment, which device class, what speed cap, what hours - plus revocable. The same layer serves wheelchairs, walking frames and prams; the robot is simply what exposed the problem first.

The Cost of the Right-of-Way Rule

Four interventions compared on one basis



Speed does nothing

Raising the cap from 15 to 25 km/h serves not one additional facility and grows the 15-minute network by only 35%. Speed is not this corridor's constraint; the shape of the network is. Any recommendation built on relaxing the speed limit is void against this data.

The strongest lever costs nothing to build

Permitting low-speed devices on footways (steps excluded) grows the 15-minute network from 20,341 m to 132,098 m, a factor of 6.5. For the same doors a person on foot faces a median terminal gap of 82.8 m against the robot's 249.6 m. The robot fails not because the AI is not capable enough, but because it is only permitted on cycle lanes.

The sequence is computed, not asserted

Change the rule first (no construction, 6.5x), then close the links (596 m, 1.79x), then do the terminals (1,841 m, bringing in the last four). Only the terminal works make all five servable. The footway figures are a lower bound: OSM footway mapping is fragmented, so real connectivity can only be better.

Dispatch Ledger: 30 Tasks, 3 Successes

Every outcome decided by the measured graph, not synthetic trajectories

Round	Right-of-way state	Reachable	Delivered	Median gap	Worst gap
A	Cycle lanes as measured	2 / 10	0 / 10	230 m	652 m
B	**Plus proposed 596 m of links**	**6 / 10**	**1 / 10**	**230 m**	**652 m**
C	Counterfactual: footways permitted	2 / 10	2 / 10	76 m	188 m

Row B is the useful one

The 596 m of new links raises network reachability from 2 to 6 but deliveries only from 0 to 1, and the median terminal gap does not move at all. The construction lands, and most tasks are still blocked at the door rather than on the network - the same conclusion the connectivity analysis reached independently.

Audit completeness 30%, the ugliest number this package records about itself

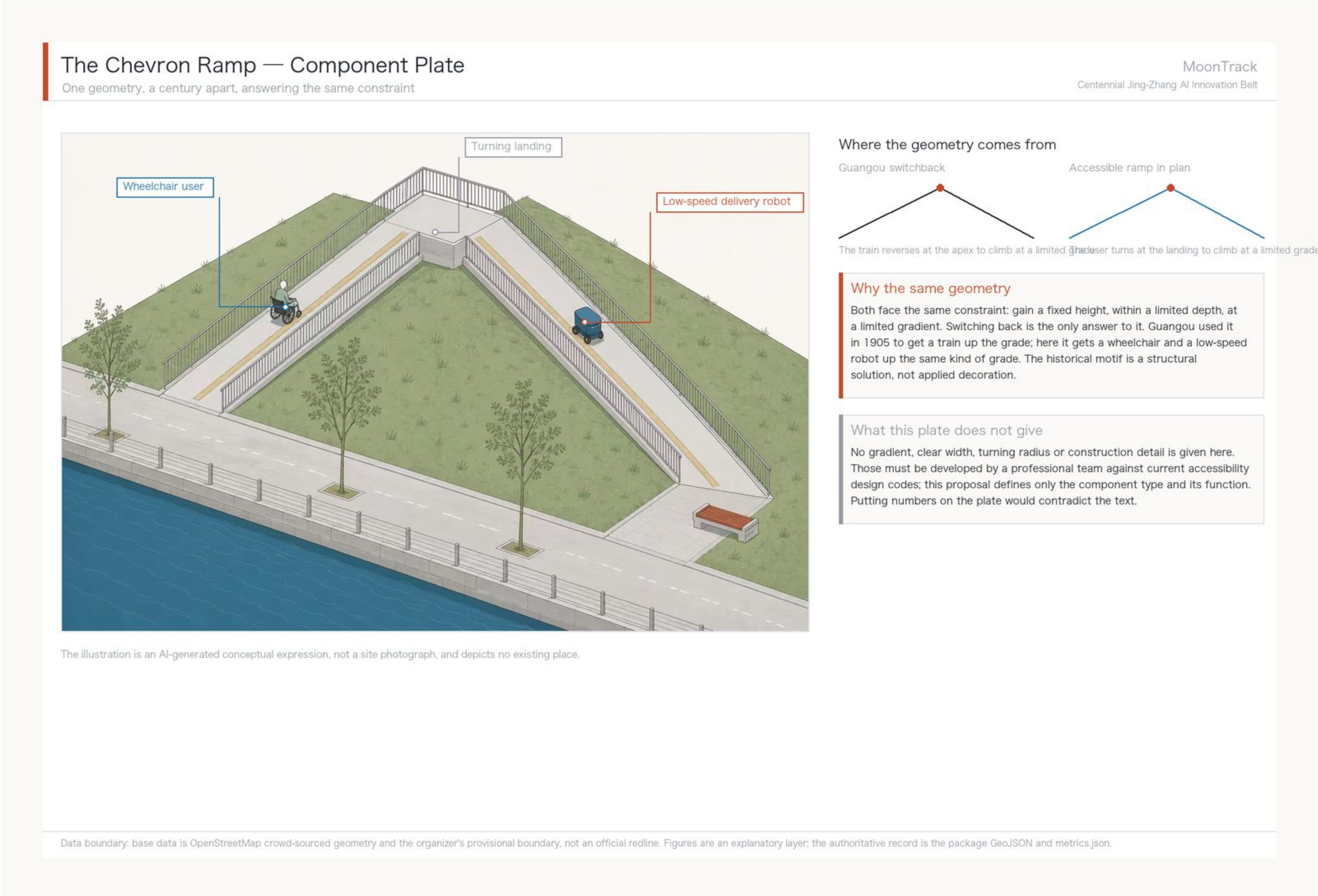
Two things drive it: rounds B and C depend on right-of-way that does not exist today, and for universities and industry anchors this proposal names no human handover party at all. The five elderly-care facilities have one, written into scenario card 02. The other five do not. That is a real operational gap, recorded in the ledger rather than left to hide behind the phrase 'the scenario is deliverable'.

No battery parameter enters the energy budget

Budget = straight-line distance x 1.5 x unit rate; consumption = legal route length x the same rate. The rate cancels, so '5 tasks over budget' is exactly the count of tasks whose legal route exceeds 1.5x the straight line - a purely geometric quantity. No claim is made about any vehicle's range.

Component 1: Chevron Ramp

The historical motif is a structural solution, not ornament



Why the same geometry

The Guangou switchback and an accessible ramp face one constraint: climbing a fixed rise at a limited gradient within a limited depth. Switching back is the only answer, which is why they take the same form.

Who it serves at once

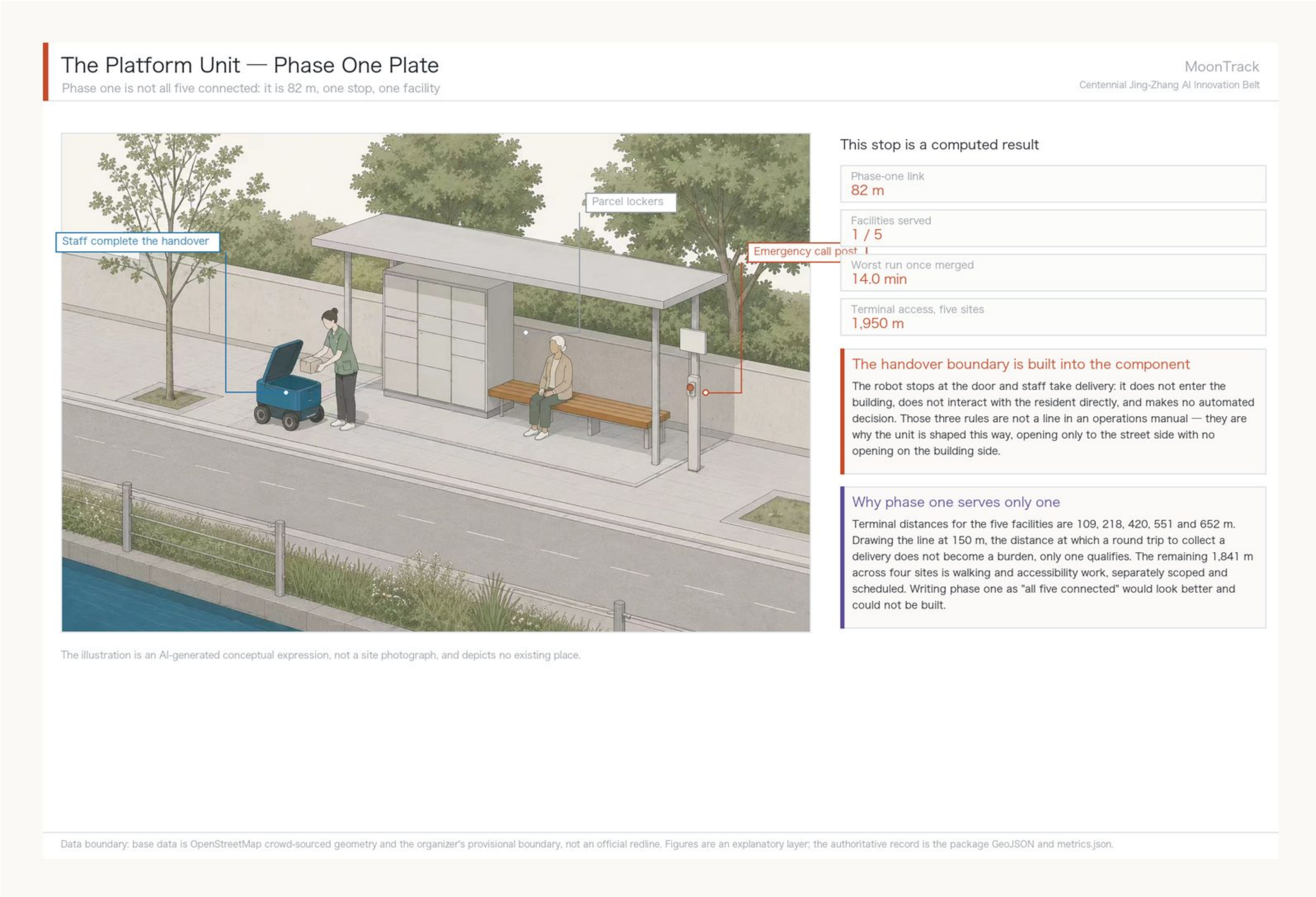
Wheelchair users and low-speed delivery robots are bound by the same gradient limit, so one ramp answers both. If the robots leave, the ramp still serves wheelchairs, prams and luggage. This is where the reversibility principle lands.

What this sheet does not give

Gradient, clear width, turning radius and structure are not specified here; they belong to a professional team. The sheet carries only figures this proposal measured and rules its text already states. The illustration is an AI-generated concept image, not a photograph.

Component 2: Platform Unit

The phase-one deliverable; the handover boundary drives the form



Phase one serves one facility, and that is computed

Phase one: 82 m of link, one facility of five, 14.0 minutes to the furthest once joined. The other four have 1,841 m of terminal gap between them, past the 150 m threshold, so they belong to walking-and-cycling works rather than equipment procurement.

The boundary is not a note, it is the form driver

No entry indoors, no contact with the resident, no automated decisions. The cabinet therefore opens only to the street and has no indoor-side aperture: staff take receipt on the street side, and the robot has no physical means of entering.

Reuse, not new build

All six robot scenario cards use existing facilities; none requires a dedicated new structure. The platform unit is a demountable module: when the pilot ends it comes out, leaving no permanent occupation of the footway.

Component 3: Observation Point

Where explainability stops being a principle and becomes a requirement

The Observation Point — Explainability Plate

Open to watch is not open data: oversight starts with being able to see what is happening

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The illustration is an AI-generated conceptual expression, not a site photograph, and depicts no existing place.

Three things the panel must answer

1. What it is doing
2. How the data is handled
3. Who supervises it

Where the three signage principles land

Signage for AI installations cannot simply state a name. The explainability-first principle becomes a concrete requirement here: the panel must answer the three questions above. The observation point itself collects no visitor data — open to watch is not open data. Entry is ramped with no steps, on the same accessibility baseline as the chevron ramp.

Data boundary: base data is OpenStreetMap crowd-sourced geometry and the organizer's provisional boundary, not an official redline. Figures are an explanatory layer; the authoritative record is the package GeoJSON and metrics.json.

The sign must answer three questions

What is it doing; how is data handled; who supervises it. A device missing any of the three does not enter a public-space pilot. It is the only hard requirement in the wayfinding system.

What is not collected

No faces, no individual movement traces. An observation point is where people watch robots, not where robots watch people: that directional distinction sets its orientation, seating and sightlines.

Why it needs a place of its own

Public acceptance is a risk shared by all ten scenario cards. Putting the robots somewhere people can stop, ask and object works better than explaining in a brochure. The illustration is an AI-generated concept image, not a photograph.

Renewal Project List and Phasing

JZ-07 is the smallest and earliest item

ID	Project	Phase	Type
JZ-07	**Field verification of three key network gaps**	**Near term**	**Mobility**
JZ-01	Heritage-park slow-mobility stitching	Near term	Public space
JZ-02	Zhongzhiyuan Qinghe innovation interface	Medium term	Blue-green
JZ-03	Campus-adjacent translation street	Medium term	Renewal
JZ-04	Dazhongsi four-quadrant pedestrian links	Medium term	Transit
JZ-05	AI service and edge-compute nodes	Long term	Infrastructure
JZ-06	Global AI week public route	Ongoing	Operations

Why JZ-07 comes first

Measurement shows that 56 m of connection across three points could grow the usable robot network from 27,560 m to 57,770 m, a return no other item on the list approaches. It must nevertheless begin with field verification: whether the breaks are mapping artefacts, whether railings or grade changes intervene, and whether kerb gradient and clear width permit passage can only be judged on site. It proceeds only if verification confirms the breaks, and is struck from the list otherwise. Almost all of its cost is verification, not construction.